

RE07 Global Infrastructure Investment and Management

Week 7

Sustainability, ESG and Emerging Trends

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Guest session: Central Bank of the United Arab Emirates

What this week does

The module has so far treated infrastructure as an asset with a contract attached. This week asks what happens when the physical world changes underneath the contract, and when policy changes above it. The organising question is not whether sustainability matters but whether it is priced, because a risk that is real and unpriced is an investment problem before it is a policy one.

A	What ESG means for an infrastructure asset	30 min
B	Physical climate risk, and whether it is priced	40 min
C	Transition risk: policy, technology and litigation	35 min
D	Green finance instruments and what they are worth	35 min
E	Measurement and disclosure, and where they fail	35 min
F	Spatial finance and asset-level analysis	35 min
G	Digital infrastructure, smart cities and cross-vertical strategy	30 min

What you already have

Where this week draws on modules you have taken, or are taking now. Nothing below is assumed: each idea is stated in full before the module is named.

Your programme	Where it was taught	What it gives this week
Real Estate Finance	RE02 Lent lecture 1, investing in the transition to net zero, and lecture 2, climate change risk. Lent lecture 4, valuation with artificial intelligence, inputs and testing.	Physical and transition risk and climate-adjusted asset pricing, in almost the same terms this week needs, and the machine learning valuation material that block F builds on.
Planning, Growth and Regeneration	PGR05 lecture 5, place-based policy and the green transition, and lecture 7, policies for whom. PGR01 Lent lecture 13, planning for net zero, and lecture 14, environmental regulation and development. PGR02 Lent week 8, the socio-ecological transition. PGR07 lecture 7, shrinking cities and left-behind places.	The spatial politics of the transition, the planning instruments that deliver it, and the question of stranded assets in places that are losing population.
Environmental Policy	EP06 throughout, particularly lectures 2 to 6 on resources, technology and innovation economics. EP02, instrument choice I to III, and regulation and innovation. EP01 Lent weeks 3, 4 and 7, the Paris Agreement, carbon markets and climate finance, and climate litigation. EP07 week 8, private governance.	This is your week. You have the science, the instruments and the law; what the module adds is what each is worth to a credit committee.
Everyone	RM03 lecture 6, spatial data mining and machine learning; lecture 3, GIS and spatial data processing; lecture 2, data types and collection.	Spatial finance is geospatial analysis applied to financial exposure. Block F is RM03 with a balance sheet attached.

Block A

What ESG means for an infrastructure asset

Three letters that behave quite differently, and the instruments behind them.

Indicative length 30 minutes

Separate the three letters, because they behave differently

Environmental factors for infrastructure are largely physical and measurable: emissions, water use, land take, biodiversity impact, and exposure to hazard. Social factors are largely about process and consent: land acquisition, resettlement, labour standards, community impact, and who bears the burden of a project whose benefits fall elsewhere. Governance is about the structure through which decisions are taken and money moves, and is the letter with the longest track record of affecting returns.

Collapsing the three into one score is what makes ESG ratings so widely criticised. For infrastructure the useful move is the opposite: treat each as a distinct risk with its own measurement problem and its own price.

The instruments that actually exist

Lender safeguards

The IFC Performance Standards and the Equator Principles, applied as conditions of finance. The most consequential instruments in practice because they bind.

Asset rating systems

Envision in North America and Infrastructure Sustainability in Australia, scoring a project against defined credits. Voluntary, and used in procurement.

Fund manager toolkits

The CDC toolkit reproduced in Weber's appendices, giving managers a process for assessing and monitoring portfolio companies.

Disclosure frameworks

Climate-related financial disclosure, now converging into accounting standards, requiring governance, strategy, risk management and metrics.

Taxonomies

Definitions of what counts as sustainable activity, used to determine eligibility for labelled instruments.

National policy instruments

Carbon prices, standards and subsidies, which are the instruments EP02 analyses and which determine the economics everything else describes.

Why lender safeguards matter more than ratings

A voluntary rating changes reputation. A lender safeguard changes whether the money arrives. Where a project cannot reach financial close without an environmental and social impact assessment, a resettlement plan and a stakeholder engagement process meeting a defined standard, those requirements are effectively regulation, imposed by contract and enforced by the threat of withdrawal.

This is private governance in the sense EP07 uses, and it has a striking property: it binds in jurisdictions where domestic regulation does not. A project in a country with weak environmental law can be held to a higher standard by its lenders than by its government, which raises an obvious question about legitimacy that the environmental policy students will want to pursue.

The social letter, for infrastructure, is mostly about land and consent

For a linear or large-footprint project the dominant social issue is land: acquisition, compensation, and where people have to be moved, resettlement. The IFC standard on land acquisition and involuntary resettlement is among the most demanding requirements in project finance and among the most frequently breached.

Beyond land sit labour standards during construction, community health and safety, and the distributional question of who bears the disturbance of a project whose benefits accrue elsewhere. Week 2 met that as planning conflict; here it is a financing condition. Projects stopped by social opposition substantially outnumber projects stopped by technical failure.

Governance has the longest track record of affecting returns

Long before climate entered investment analysis, governance did: board composition, minority protection, related-party transactions, disclosure quality and the alignment of managers with owners. For infrastructure the governance questions are specific and consequential.

Who controls a consortium and how are deadlocks resolved. What are the related-party arrangements where a sponsor is also the contractor or the operator. How is the asset governed after a fund exits. And in emerging markets, what is the integrity risk in a sector characterised by large contracts, discretionary approvals and long relationships. Of the three letters this is the one with the clearest evidence of a financial consequence.

Block A discussion

Seminar questions

1. Which of the three letters has the clearest link to financial performance for an infrastructure asset, and what is the evidence?
2. A lender safeguard binds where domestic law does not. Is that a good thing? Who is accountable for the standard?
3. Why do ESG ratings for the same company differ so much between providers, and what does that imply about using them in an investment process?

Block B

Physical climate risk, and whether it is priced

Immobile, long-lived assets in a changing hazard distribution.

Indicative length 40 minutes

Infrastructure is the asset class where physical risk bites hardest

Three properties combine badly. The asset cannot be moved, so exposure is fixed at the moment of siting. Its life is measured in decades, so it will experience a hazard distribution materially different from the one it was designed for. And its revenue often depends on being available continuously, so a disruption is not merely a repair cost but a revenue loss and possibly a contractual failure.

Against that, the design standards, the insurance and the force majeure drafting were all set against the historic distribution. That mismatch is the subject of the block and it is a pricing question before it is an environmental one.

Physical risk is priced, at long maturities

23.4bp

Additional annualised issuance cost for long-term municipal bonds associated with a one per cent increase in climate risk exposure.

Painter examines United States municipal bonds and finds that counties with higher climate risk pay more to issue, through both underwriting fees and interest rates, and that the effect is concentrated in long-dated issues.

The maturity pattern is the interesting part. Short-dated paper shows little effect and long-dated paper shows a clear one, which is consistent with the market pricing a hazard that materialises over decades rather than years.

For infrastructure this is close to a direct result. Municipal issuance funds local infrastructure, the maturities are comparable, and the finding says that the market does price physical exposure where the tenor is long enough for it to matter.

What investors say they believe

Already

Institutional investors reporting that climate risks, and regulatory risks in particular, have begun to materialise in their portfolios.

Krueger, Sautner and Starks surveyed institutional investors on climate risk perceptions. Investors believe climate risks have financial implications for portfolio firms and that some have already begun to materialise, with regulatory risk most prominent.

On what to do about it, the larger, longer-horizon and more ESG-oriented investors favour risk management and engagement over divestment, which is a considered position rather than a soft one: selling an asset changes who owns the risk without changing the risk.

Set this against Painter. One paper says the market prices the hazard; the other says investors say they are watching it. Neither establishes that the pricing is correct in magnitude, only that it is not zero.

Where physical risk enters an infrastructure transaction

- Siting and design, where the standard is set against a historic return period that may no longer describe the hazard.
- Construction, where extreme weather is a delay event and the allocation of that delay is a contractual question.
- Operation, through availability, damage and the cost of adaptation works that were not in the original capital plan.
- Insurance, where cover is repriced annually against a twenty-five year exposure, and withdrawal is a live possibility in some geographies.
- Refinancing, where a lender in year ten assesses a hazard distribution that has moved since financial close.
- Terminal value, where a buyer in year twenty-five prices what remains of the asset's life under conditions nobody modelled.

The wedge between intent and action

A recurring finding in the behavioural literature on climate adaptation is that intention forms while a hazard is salient and decays before it is acted on. Households and firms report an intention to adapt after a flood and largely do not follow through once the water has receded.

The implication for asset owners is uncomfortable. If adaptation decisions are made under salience rather than under expected value, then the timing of adaptation investment will be wrong even where the analysis is right, and capital expenditure plans built on stated intent will overstate what happens. This is a live research area and it connects directly to work being done in this department.

Acute and chronic hazards behave differently

Type	Examples	How it hits an asset	What it does to finance
Acute	Flood, storm, wildfire, extreme heat event	Damage, outage, and a claim	Insurance cost, deductibles, and a bad year in the ratio
Chronic	Sea-level rise, mean temperature, water stress, subsidence	Gradual loss of capacity, rising maintenance, eventual unviability	Terminal value, refinancing, and the availability of insurance at all
Compound	Heat with drought, storm surge on a rising baseline	Correlated failures across a portfolio at once	Diversification fails exactly when it is needed

Adaptation is an investment decision with a timing problem

Adapting an asset costs money now to avoid a loss later, which makes it an ordinary present value calculation with two awkward features. The loss avoided is uncertain in timing and magnitude, and the discount rate applied to a sixty-year horizon determines the answer, as week 2 established.

There is also an option value argument that cuts both ways. Waiting preserves the option to adapt with better information, which favours delay. But some adaptations are far cheaper if designed in at the outset than retrofitted, which favours acting now. The practical resolution is adaptive design: build in the capacity to adapt later at low cost, which is a real option in the RE01 sense and one of the clearest applications of that material in the module.

Insurance retreat is the near-term transmission channel

The most immediate financial effect of physical climate risk is not damage but the price and availability of cover. Insurers reprice annually, and in the most exposed geographies have withdrawn from lines altogether.

For an infrastructure asset that is a serious problem, because insurance is not optional: finance documents require specified cover, and an inability to maintain it can be an event of default. An asset that becomes uninsurable becomes unfinanceable, and it does so long before it becomes physically unusable. That transmission channel deserves more attention than it receives and it is a good dissertation topic.

Block B discussion

Seminar questions

1. Painter finds physical risk priced at long maturities and not short. Is that evidence of an efficient market or of a market that only prices what it must?
2. An insurer reprices annually and an asset lasts twenty-five years. Who is carrying the tail, and what would you pay to remove it?
3. If investors prefer engagement to divestment, what does engagement actually consist of for an infrastructure asset held through a fund?

Block C

Transition risk

Policy, technology and litigation, and which of the three actually moves a valuation.

Indicative length 35 minutes

Three channels, three time constants

Channel	How it arrives	Speed	Who is exposed
Policy	Carbon pricing, standards, phase-out dates, subsidy withdrawal	Years, and discontinuous	Emitting assets, and assets built on a subsidy
Technology	Cost declines that make an incumbent asset uneconomic	A decade, and then suddenly	Assets with long lives in a fast-moving cost curve
Litigation	Challenge to consents, to policy statements, and to directors	Unpredictable, and increasingly rapid	Consented but unbuilt projects, and their sponsors
Market	Customer and investor preference shifting demand	Gradual, and self-reinforcing	Assets whose offtakers face their own transition

Technology risk is asymmetric for long-lived assets



EP06 teaches the economics of technology cost decline: learning by doing, economies of scale, knowledge spillovers, and the resulting experience curves along which the cost of a technology falls with cumulative deployment.

For an infrastructure investor that literature has a specific implication. An asset with a thirty-year life competing against a technology on a steep experience curve faces an opponent that gets cheaper every year while the asset does not. Generation assets are the obvious case, and the same logic now applies to any asset whose competitor is improving, including transport modes and parts of the digital estate. The asymmetry is that the incumbent cannot get cheaper at the same rate, because its cost is largely sunk.

Litigation has become a transition risk channel



Climate litigation has expanded from challenges to individual consents into challenges to national policy, to policy statements underpinning consenting regimes, and to the conduct of companies and their directors. EP01 devotes a Lent lecture to it.

For infrastructure the exposure is specific. A consented project can lose its consent if the policy behind it is quashed, as the Heathrow case in week 2 demonstrated in both directions. That risk sits after the sponsor has spent heavily on development and before construction has begun, which is the most expensive point in the life of a project at which to face a delay.

What a stranded asset actually is



- An asset that suffers premature write-down, devaluation or conversion to a liability because of a change in the environment it operates in.
- The change may be regulatory, technological, or a shift in demand, and it is the earliness that defines the concept rather than the loss itself.
- For infrastructure the risk is concentrated in assets whose revenue depends on an activity being phased out, and in assets in places losing population.
- The financial consequence depends on who holds the asset at the moment of stranding, which is a question about contract expiry and exit timing as much as about the asset.
- A concession expiring before the stranding event transfers the problem to the state, which is a risk allocation question the contract should have addressed and usually has not.

What a carbon price does to an infrastructure asset



A carbon price raises the operating cost of an emitting asset and lowers the relative cost of its clean competitor. For an asset with contracted revenue and no pass-through, that is a direct margin compression. For an asset that can pass it through, it becomes a demand question.

EP02 supplies the instrument analysis and the reason a price is generally preferred to a standard: it equalises marginal abatement cost across sources and lets the market find the cheapest reductions. What the investment perspective adds is that a price announced today and rising on a stated path is a very different risk from a price that may be introduced at an unknown level, and much of the transition risk in real portfolios is uncertainty about the instrument rather than the instrument itself.

The architecture the instruments sit inside



- The Paris Agreement establishes a bottom-up structure of nationally determined contributions with a ratchet, rather than a top-down allocation.
- That structure means transition risk is national and heterogeneous, which for a global portfolio is a jurisdiction-by-jurisdiction problem rather than a single scenario.
- Carbon markets and finance provisions sit within it, and EP01 devotes a Lent lecture to each.
- Litigation increasingly uses the architecture as a benchmark: a policy inconsistent with a stated commitment is challengeable.
- For this course the practical consequence is that a national commitment is a forward-looking statement about future policy risk, and can be read as such by an investor.

Block C discussion

Seminar questions

1. Rank the four transition channels by how much they would move the valuation of a gas-fired power station with a fifteen-year contracted offtake.
2. A concession expires five years before an expected phase-out date. Who bears the stranding risk, and what should the contract have said?
3. Litigation risk cannot be insured or easily priced. How should a sponsor account for it in an investment decision?

Block D

Green finance instruments and what they are worth

Labels, premia, and the difference between signalling and subsidy.

Indicative length 35 minutes

What a labelled instrument actually is

A green bond is an ordinary bond whose proceeds are earmarked for defined categories of expenditure, with reporting on allocation and often on impact, and usually with an external opinion confirming that the framework meets a standard. The credit risk is the issuer's, unchanged by the label.

That last point does most of the analytical work. If the claim on the issuer is identical, then any difference in yield must come from investor preference rather than from risk, which is exactly what makes the green bond literature such a clean test of whether investors will pay for environmental characteristics.

How much investors will pay for the label

About 2bp

Estimated yield differential between a green bond and a matched conventional bond, over the period studied.

Zerbib matches green bonds to counterfactual conventional bonds and estimates the yield differential in a two-step procedure. The premium is negative and small: the green bond yields about two basis points less than its conventional counterpart on average, with the effect more pronounced for financial and lower-rated issuers.

Two basis points is a real effect and it is not a subsidy of any consequence. An issuer choosing between labelled and unlabelled issuance is not making a financing decision of material economic significance, which means the reasons for issuing green must lie elsewhere: investor diversification, signalling, and internal discipline.

Students should be able to state this without either dismissing green finance or overstating what it delivers.

What issuance actually signals

Positive

Stock market reaction to the announcement of a corporate green bond issue, in Flammer's sample.

Flammer finds that investors respond positively to green bond announcements, more strongly for first-time issuers and for bonds certified by a third party. Issuers subsequently improve their environmental performance and attract more long-term and green investors.

Her interpretation is signalling: issuing a green bond is a costly and public commitment, and the market reads it as credible information about intentions rather than as a change in the cost of capital.

The certification finding matters here. The effect is stronger where a third party has verified the framework, which is the week 5 certification mechanism in a different market.

Two readings of the green bond evidence

The optimistic reading

- Announcement returns are positive and environmental performance improves afterwards
- Certification strengthens the effect, so the market distinguishes credible commitments from cheap ones
- The instrument has created an investor base that did not exist a decade ago
- Earmarking imposes internal discipline on which projects are pursued

The sceptical reading

- A two basis point premium is not a meaningful change in the cost of capital
- Proceeds are fungible in practice, and earmarking may relabel rather than redirect
- Improvement after issuance may reflect firms that were already improving choosing to issue
- The instrument grows fastest where the standard is weakest, which is the usual pattern with labels

The labelled instrument family

Instrument	What is committed	Where it fits infrastructure
Green bond	Proceeds are allocated to eligible projects	Well suited: a project has identifiable eligible expenditure
Social and sustainability bond	Proceeds to social or mixed categories	Social infrastructure, and access-oriented projects
Sustainability-linked bond or loan	The coupon or margin varies with performance against targets	Suits an operating business better than a single project
Transition instrument	Proceeds or targets tied to decarbonising a high-emitting activity	Contested, and where the definitional argument is hardest
Blue and resilience instruments	Water, marine, or adaptation-focused categories	Emerging, small, and increasingly relevant to adaptation capital expenditure

Taxonomies decide what counts, and that is a political act

A taxonomy defines which activities qualify as sustainable, usually with technical screening criteria and a do-no-significant-harm test. Eligibility under it determines access to labelled instruments and, increasingly, to regulatory advantages.

Because eligibility has value, the definitions are contested by the industries being classified, and the arguments over whether particular energy sources qualify have been conducted at the highest political level. For students the lesson is a general one about instrument design: whenever a definition confers an economic benefit, the definition becomes the object of lobbying, and the technical criteria are where that lobbying lands.

Greenwashing, and what actually constrains it

- A label with value and weak verification invites misuse, which is the standard prediction and it has been borne out.
- The constraints that appear to work are external review of the framework, mandatory reporting on allocation, and supervisory attention to marketing claims.
- The constraint that does not work is investor scrutiny alone, because the investor buying a labelled instrument has already selected on wanting the label.
- Flammer's finding that certified issues get a stronger market reaction suggests the market does distinguish, imperfectly.
- This is the week 5 verification problem again: the assurance is only as good as the incentives of the party providing it.

Block D discussion

Seminar questions

1. If the green premium is two basis points, why does anyone issue a green bond? Answer without using the word reputation.
2. Flammer finds certification strengthens the market reaction. Who should pay the certifier, and what does week 5 say about that arrangement?
3. Would a larger premium be a good thing? What would it imply about how investors were pricing the conventional bond?

Block E

Measurement and disclosure

What the frameworks require, what is actually measurable, and where the gap is.

Indicative length 35 minutes

What a modern disclosure framework asks for

- Governance: who on the board is responsible, and how the responsibility is exercised.
- Strategy: which risks and opportunities have been identified, over what horizons, and how they affect the business plan.
- Risk management: how climate risks are identified, assessed and integrated with other risks.
- Metrics and targets: what is measured, against what target, with what methodology.
- Scenario analysis: how the business performs under defined warming and policy pathways, which is the requirement organisations find hardest.
- The structure is deliberately the same as any other risk disclosure, which is the point: climate is being brought into an existing apparatus rather than given a separate one.

The measurement problem, stated precisely

Disclosure requires numbers. For an infrastructure portfolio the numbers that matter are asset-level: where each asset is, what hazard it faces, what it emits, and how those change under a given pathway.

Almost none of that is available from conventional sources. Company reporting is aggregated to entity level, which averages away the location that determines physical exposure. Emissions are self-reported, with wide latitude in boundary and method. And for unlisted assets held through funds, an investor frequently does not know precisely what it owns at asset level, let alone where each asset sits. That gap is what the next block is about.

Where disclosure breaks down

Boundary

What is in scope. Emissions from an asset, from its supply chain, or from what it enables, and the third is where the largest numbers and the loudest arguments are.

Aggregation

Entity-level reporting averages away the location that determines physical exposure, which is precisely what an investor needs.

Self-reporting

The party being assessed produces the data, which is the verification problem from week 5 in a new setting.

Comparability

Different methods produce different numbers for the same asset, which is why ratings from different providers diverge so widely.

Forward-looking data

Scenario analysis requires a hazard projection at asset level, which most organisations cannot produce and few disclose.

Coverage

Unlisted, private and emerging market assets, which is much of the world's infrastructure, are the least covered by every framework.

Block E discussion

Seminar questions

1. Disclosure frameworks require scenario analysis. What would a credible scenario analysis for a single infrastructure asset actually contain?
2. Emissions are self-reported and verified by a party paid by the reporter. Where have you seen that structure before, and did it work?
3. If unlisted and emerging market assets are least covered, what does aggregate portfolio disclosure actually tell an investor?

Block F

Spatial finance

Geospatial data, machine learning, and asset-level risk for assets that cannot move.

Indicative length 35 minutes

The definition

Spatial finance

The integration of geospatial data and analysis into financial theory and practice.

Caldecott, McCarten, Christiaen and Hickey define the field and set out why it has become possible now: new satellite constellations providing frequent, high-resolution earth observation, machine learning methods able to extract features from that imagery at scale, and cloud computing making the processing economic.

The consequence is that an analyst can, in principle, identify physical assets, attribute them to owners, and assess their characteristics and exposures independently of what the owner chooses to report.

For infrastructure this is not a marginal improvement. It is the difference between an entity-level average and an asset-level measurement, and it addresses precisely the aggregation and self-reporting failures from block E.

Why infrastructure is the natural application

- The assets are physically fixed, so location is a permanent characteristic rather than a snapshot.
- They are large and visible, which makes them detectable from imagery in a way that most corporate assets are not.
- Their exposure is determined almost entirely by where they are, since a hazard map and a coordinate give most of the answer.
- Their lives are long enough that the hazard distribution changes materially within the holding period.
- And ownership is often opaque, held through funds and holding companies, so an independent route to asset-level information has real value.

What the method can and cannot do

Capability	Maturity	Caveat
Locate and identify large physical assets	Established	Attribution to an owner is the hard part, not detection
Assess physical hazard exposure at a coordinate	Established	Hazard models disagree, and downscaling introduces error
Estimate emissions from observable proxies	Developing	Works for some sectors and poorly for others
Monitor construction and operation over time	Established	Frequency and cloud cover constrain what can be observed
Attribute assets to corporate owners	Developing	Requires ownership data that is often private
Translate exposure into a financial number	Immature	The damage function is the weakest link in the whole chain

The damage function is where the chain is weakest

A complete analysis runs from hazard to exposure to vulnerability to financial consequence. The first two links are increasingly well served by data. The third, how much damage a given hazard does to a given asset, rests on engineering assumptions with wide uncertainty. The fourth, what that damage is worth once insurance, contract and adaptation are accounted for, is barely modelled at all.

The practical implication is that spatial finance currently produces excellent exposure data and weak valuation conclusions. That is a reason to use it for screening and for engagement rather than for precise repricing, and it is an honest thing to say to students who will be sold the opposite.

Questions to put to the guest

- Where in the chain from hazard to financial consequence is the largest source of error, and what would reduce it fastest?
- What can a central bank do with asset-level exposure data that a private investor cannot?
- Physical risk appears to be priced at long maturities and not short. Is that efficiency, or a market pricing only what it is forced to?
- What would it take for asset-level climate data to enter a credit rating rather than sitting alongside it?
- Where is the gap between what is technically measurable and what is actually used in decisions, and what closes it?

A worked screening exercise, in outline

Take a single infrastructure asset. Establish its coordinates from public sources. Overlay a flood hazard layer at the relevant return periods and a heat or water stress layer as appropriate. Record what the asset's design standard was, from the environmental statement or the company's disclosure. Compare the two. Then ask what the finance documents say about availability, insurance and force majeure, and whether the mismatch you have found would trigger any of them.

The output is not a valuation. It is a statement of exposure, an identification of which contractual provisions are engaged, and a list of questions to put to the owner. That is a realistic account of what spatial finance can currently deliver, and it is a genuinely useful piece of analysis.

Data sources students can actually use

- National hazard mapping, which for flood is publicly available at high resolution in many countries.
- Open earth observation programmes, which provide free imagery at moderate resolution and frequent revisit.
- Open geospatial datasets of infrastructure assets, which are incomplete and improving.
- Company environmental statements and planning applications, which contain design standards and site-specific assessment.
- Regulatory disclosure, which is entity level but sometimes identifies major assets.
- The binding constraint is almost never the hazard data. It is knowing precisely which assets an owner holds and where they are.

Block F discussion

Seminar questions

1. Spatial finance produces good exposure data and weak valuation conclusions. Is it useful to an investor today, and for what specifically?
2. If an analyst can measure an asset's emissions independently of what its owner reports, what happens to the disclosure regime?
3. Design an asset-level exercise for a single infrastructure asset using publicly available geospatial data. What would you be able to say, and what not?

Block G

Digital infrastructure, smart cities and cross-vertical strategy

The newest members of the asset class, and the least tested.

Indicative length 30 minutes

Digital public infrastructure is a category worth taking seriously

Abor's chapter on digital public infrastructure describes a layer that is genuinely infrastructural in the economic sense: identity, payments and data exchange systems that other services are built on top of, exhibiting network effects, requiring interoperability, and raising the classic questions about access, monopoly and governance.

It is not what infrastructure funds mean by digital infrastructure, which is towers, fibre and data centres. Both are worth studying and confusing them is a common error. The first is a public utility question; the second is an asset class question with an unusually short technology cycle.

Data centres, and why lenders find them difficult

- They are contracted to strong tenants, often with long leases, which looks like contracted infrastructure revenue.
- The tenant base is highly concentrated, which is a credit exposure a utility lender would never accept.
- The technology cycle, in cooling, chip density and power requirements, is far shorter than the debt tenor.
- Power availability has become the binding constraint on siting, which links the sector directly to the grid investment question.
- And the residual value at the end of a lease depends on assumptions about a technology nobody can forecast for twenty years.
- Whether these belong in an infrastructure portfolio is the clearest live example of the boundary problem from week 1.

Smart cities, treated analytically

Kummitha and Crutzen trace the concept through its variants, distinguishing a technology-led reading, in which information technology is deployed to make urban systems more efficient, from a human-led reading, in which governance, human capital and social inclusion are the objects and technology is instrumental.

The distinction matters for investment because the two readings imply different assets. The technology-led reading implies procurable systems with identifiable vendors and revenue models. The human-led reading implies institutional reform, which nobody can invest in. Much of the disappointment with smart city programmes comes from procuring the first while claiming the second.

Machine learning in infrastructure credit

- Abor's final chapter applies machine learning models to the credit risk assessment of infrastructure bonds, comparing model performance in and out of sample and examining which features matter.
- The appeal is that credit assessment for a heterogeneous asset class with few observations is exactly where flexible functional forms might help.
- The difficulty is the same as everywhere else: a model that fits well out of sample on historic data has still only seen the regimes contained in that data.
- For infrastructure that limitation is severe, because the interesting events are rare, and the sample is dominated by a period of falling rates and stable policy.
- RE02 covers the same tension for valuation in its Lent lecture on artificial intelligence, inputs and testing.

Where this block runs out of evidence

Cross-vertical strategy, meaning the deliberate integration of energy, transport, digital and water systems into a single investment or operating proposition, is discussed constantly and studied almost not at all. There is no comparator course teaching it and no substantial literature to set.

That is worth stating to students rather than concealing with confident-sounding generalities. Where a field is genuinely immature, the appropriate teaching move is to say so, give the best available thinking, and point out that this is where a dissertation could contribute.

The transition is an infrastructure investment programme

Whatever else it is, decarbonisation is a very large infrastructure programme: generation, transmission and distribution, storage, charging, heat networks, hydrogen where it works, and the industrial plant behind all of it. Everything this module has taught applies to it directly.

That framing is useful because it converts an argument about ambition into a set of familiar questions. What is the revenue model. Who bears demand risk. Is there a pipeline. What is the consenting timetable. Who supplies the capital and at what tenor. Students who can ask those questions of a national decarbonisation plan will ask better questions than most of the people writing them.

The grid is the binding constraint, and it is a regulated asset

- Generation can be built faster than the network to connect it, and connection queues now run to years in several markets.
- Network investment is set by a price control, which means the pace of the transition is partly determined by a regulatory determination.
- That is week 2 material with a week 7 consequence: an allowed return and a capital allowance are climate policy instruments whether or not anyone calls them that.
- Anticipatory investment, building ahead of confirmed demand, is exactly what a regulator is designed to be sceptical of.
- Which means the regulatory framework that was built to protect consumers from over-investment is now a constraint on the investment the transition requires.

Block G discussion

Seminar questions

1. Should a hyperscale data centre platform sit in an infrastructure portfolio? Answer using the six properties from week 1.
2. Kummitha and Crutzen distinguish technology-led from human-led smart cities. Which can be invested in, and what does that imply about the disappointment with smart city programmes?
3. Machine learning credit models have only seen one interest rate regime. Does that make them useless, or useful with a caveat?

Block H

Reserve material

Slides held back from the main sequence.

Indicative length As required

Candidate cases for Week 7

Case	What it illustrates	Material
A coastal asset with sea-level exposure	Physical risk, insurance withdrawal and adaptation timing	Public hazard maps and company disclosure
A gas asset with a phase-out date	Transition risk and stranding before concession expiry	National policy and company reporting
A sovereign or utility green bond	The premium, the framework and the second-party opinion	Prospectuses and opinions are public
An offshore wind contract for difference	A policy instrument turned into a bankable revenue stream	Auction results are public
A data centre platform	The boundary of the asset class, and power as the binding constraint	Company and market reporting
A smart city programme	Technology-led against human-led, and what was procured	Municipal documentation and evaluations

An alternative running order

Where the cohort has a strong environmental policy contingent, invert the week: open with block C on transition risk and let those students supply the policy and legal content, then use blocks B and D to ask what it is worth, and close on F.

Where the cohort is mainly finance, keep the order as given but compress block A, since the instruments are best learned by application, and give the time to F, which is where the finance students have least prior exposure and most to gain.

Sources drawn on for this deck

- Weber, B., Staub-Bisang, M. and Alfen, H. W. (2016), Infrastructure as an Asset Class, chapters 1.2 and 2.2, and appendices A to D.
- Abor, J. Y. and others (eds) (2025), The Routledge Handbook of Infrastructure Finance, chapters 20, 21, 25, 33, 34 and 35.
- Delmon, J., Private Sector Investment in Infrastructure, chapter 25 sections 9 to 12.
- Flyvbjerg, B. (ed.) (2017), The Oxford Handbook of Megaproject Management, chapters 19 and 25.
- Journal articles as cited on each slide.

Where Week 7 leaves us

Physical climate risk is priced, at long maturities and not short, which is what a market pricing a slow hazard should look like.

The green label is worth about two basis points, so the reasons to use it are signalling and discipline rather than the cost of capital.

Disclosure fails at asset level, which is exactly the level at which an immobile long-lived asset's exposure is determined.

Spatial finance closes the exposure gap and not yet the valuation gap, which makes it a screening instrument rather than a pricing one.